Published in final edited form as:
Gastroenterology. 2023 August ; 165(2): 499–501.e4. doi:10.1053/j.gastro.2023.04.007.

EFFECTS OF FGF19 ANALOG ALDAFERMIN IN PATIENTS WITH BILE ACID DIARRHEA: A RANDOMIZED, PLACEBO-CONTROL TRIAL

Joelle BouSaba[1], **Monique Torres**[1], **Saam Dilmaghani**[1], **W. Scott Harmsen**[1], **Lei Ling**[2], **Michael Camilleri**[1]

[1]Clinical Enteric Neuroscience Translational and Epidemiological Research (CENTER), Mayo Clinic, Rochester, MN;

[2]NGM Biopharmaceuticals, South San Francisco, CA

Irritable bowel syndrome (IBS) is a functional gastrointestinal (GI) disorder characterized by recurrent abdominal pain and altered bowel movements[1]. Around 25–50% of patients with diarrhea-predominant IBS (IBS-D) have evidence of bile acid (BA) diarrhea (BAD)[2].

In patients with idiopathic BAD with IBS-D, there is decreased FGF19 production[3, 4]and decreased negative feedback on BA synthesis, as manifested by increased serum 7α-hydroxy-4-cholesten-3-one (7αC4). BA sequestrants are approved for the treatment of hyperlipidemia and are used off-label to treat BAD[5]. However, they might interfere with the absorption of other drugs.

Aldafermin (NGM282), a 190 amino acids peptide, is an engineered analog[6] of recombinant human FGF19 with a 95.4% homology. The aim of this study was to examine the effects of aldafermin on BA synthesis, BA excretion, and bowel function in patients with IBS-D and BAD.

We performed a 28-day, randomized, double-blinded, placebo-controlled (1:1 ratio) trial of aldafermin vs placebo in patients with BAD (Supplemental Figure, **upper panel**). This

Address for correspondence: Michael Camilleri, M.D., Mayo Clinic, Charlton Bldg., Rm. 8-110, 200 First St. S.W., Rochester, MN 55905, Telephone: 507-266-2305, camilleri.michael@mayo.edu.
Authors' contributions:
Joelle BouSaba: recruitment and medical care of participants during study; lead author
Saam Dilmaghani: recruitment and medical care of participants during study; co-authors
Monique Torres: study coordinator and recruitment of patients; co-author
W. Scott Harmsen: biostatistics; co-author
Lei Ling: coordination of pharmacokinetic measurement, NGM pharmaceuticals; co-author
Michael Camilleri: principal investigator, conceptualization of study, grant recipient, senior author

Clinical trial registration: ClinicalTrials.gov, NCT#05130047

Disclosures:
M. Camilleri: This study sponsor, NGM Biopharmaceuticals, was not involved in the recruitment of participants or analysis of data.
L. Ling: employee and stockholder of NGM Biopharmaceuticals
The other authors have no conflicts of interest.

study was conducted at Mayo Clinic between December 2021 and November 2022. Block randomization ensured equal numbers randomized to placebo and aldafermin in two blocks: patients on BA sequestrants and patients off BA sequestrants.

Patients were required to have a clinical diagnosis of IBS-D ages 18–75 years, and BMI 18.0 to 45.0 kg/m^2. Diagnosis of BAD based on at least one of these criteria: 1) serum 7αC4 ≥52ng/mL; 2) fecal BA >2337μmoles/48 hours; 3) total fecal BA >1000μmoles/48 hours + 4 % primary BA; 4) fecal primary BA >10%/48 hours.

The co-primary endpoints were the fasting serum 7αC4 from baseline on day 28 and the stool consistency averaged over the 28-day study. The latter was originally listed as a secondary outcome in the original protocol. Further details regarding experimental design, ethical approval, eligibility criteria, assessment of endpoints and statistical analysis (based on 2-group comparisons using ANCOVA) are provided in the Supplemental Material.

Among 35 patients enrolled, 31 were randomized. One patient withdrew after one dose of aldafermin due to severe nausea unresponsive to antiemetics. Three patients in the aldafermin group and 2 in the placebo group remained on a stable dose of BA sequestrants during the study. The demographics, baseline and on treatment bowel functions, BA synthesis and excretion are reported in Table 1.

Patients on aldafermin had significantly decreased serum 7αC4 at days 14 and 28 (Supplemental Figure, **lower panel**) but no significant change in stool consistency during days 1–28, 1–14 or 15–28. Aldafermin-treated group also had significantly decreased fecal total BA and percent secretory BA at days 14 and 28.

There were no significant differences in abdominal pain or stool frequency between the two groups during days 1–28, 1–14, or 15–28. However, there was numerically improved stool consistency in patients on aldafermin during days 15–28 (Bristol Stool Form Scale 4.6 for aldafermin vs 4.7 for placebo, p=0.082), particularly in week 4 of treatment (p=0.047). Total IBS quality of life (QOL) score and scores on each of the 8 QOL domains were not significantly different on day 28 in patients on aldafermin or placebo.

The aldafermin group had a greater increase in low-density lipoprotein cholesterol (LDL-C) from baseline compared to placebo group (p=0.052). Four patients on aldafermin were considered to have a clinically significant increase in LDL-C at day 28 which was normalized 1 month after cessation in 3 of the 4 patients.

There were more adverse events in the group receiving aldafermin compared to those receiving placebo (Supplemental Table). One patient who was randomized to aldafermin was hospitalized with myocardial infarction with normal coronary angiogram, an event deemed unrelated to the study drug. Over 28 days, rescue medication with loperamide was received by two participants on placebo (total 6 and 16 mg) and 3 on aldafermin (total 20 mg, 20 mg, and 12 mg).

In this study, we showed that aldafermin, 1mg SQ daily, led to a significant change in the biochemical measurements of BAD in patients with IBS-D. Improved stool consistency with aldafermin was only observed during week 4 of treatment.

While the biochemical changes associated with aldafermin have been previously demonstrated in other patient populations[6], increased frequency of bowel movements with higher doses has been reported as a side effect of aldafermin,[7, 8] and acceleration of colonic transit documented with higher dose of 6 mg NGM282 (aldafermin) was associated with increased frequency in contrast to the 1 mg dose used in the current study.[9]

Overall, aldafermin had an acceptable side effect and tolerability profile. More patients on aldafermin had a clinically but not statistically significant increase in LDL-C levels at study completion (Supplemental Table). Aldafermin has been associated with increased LDL-C in prior studies[8] secondary to decreased conversion of cholesterol to BA and this could be reversed by co-administration of rosuvastatin.[10]

The strengths of this study include assessment of the effects of aldafermin on both biochemical endpoints and patient reported outcomes; the primary biochemical endpoint was met in this small study sample. While the small sample size in our study had sufficient power to detect >1 or 1.5 points on the 7-point BSFS, such a change may not be observed with most agents approved for IBS-D, and a larger sample size would have been needed to detect a significant effects on symptoms.

Limitations include enrollment of 5/30 patients on BA sequestrants, though block randomization ensured equal distribution of patients across both arms, and inclusion of patients with stool frequency and consistency that were less severe than patients in other IBS-D trials used for approval of medications. Although this trial was conducted at a tertiary referral center, all the participants were residents of the tri-state area of Minnesota, Wisconsin, and Iowa seeking care in their communities for gastroenterological disorders.

Overall, these data suggest that this FGF19 analog is a promising approach to the treatment of BAD.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgement:

The authors thank Wassel Sannaa and Ting Zheng for recruitment and medical care of participants during study, Sanna McKinzie and Irene Busciglio for assistance in design of the protocol, Paula Carlson for biospecimen preparation and biochemical measurements, and Cindy Stanislav for secretarial assistance.

Funding support:

This research study was supported by a research grant from NGM Biopharmaceuticals. Michael Camilleri's research on bile acid diarrhea is supported by grant R01-DK115950 from National Institutes of Health. This publication (or project) was made possible by the CTSA Grant UL1-TR002377 from the National Center for Advancing Translational Sciences (NCATS), a component of the National Institutes of Health (NIH). Its contents are solely the responsibility of the authors and do not necessarily represent the official views of the National Institutes of Health.

Data transparency statement:

Only deidentified participant data will be provided as the mean and SD or median and interquartile range for each treatment group, not as individual data. All results will also be made available in ClinicalTrials.gov NCT#05130047 on publication of the paper. Additional related documents will be available, if deemed appropriate and after approval of a proposal, by request to the senior author with documentation of investigator support, and with a signed data access agreement.

REFERENCES

1. Mearin F, Lacy BE, Chang L, et al. Bowel Disorders. Gastroenterology 2016;150:1393–1407.e5.
2. Valentin N, Camilleri M, Altayar O, et al. Biomarkers for bile acid diarrhoea in functional bowel disorder with diarrhoea: a systematic review and meta-analysis. Gut 2016;65:1951–1959. [PubMed: 26347530]
3. Johnston IM, Nolan JD, Pattni SS, et al. Characterizing Factors Associated With Differences in FGF19 Blood Levels and Synthesis in Patients With Primary Bile Acid Diarrhea. Am J Gastroenterol 2016;111:423–432. [PubMed: 26856750]
4. Walters JR, Tasleem AM, Omer OS, et al. A new mechanism for bile acid diarrhea: defective feedback inhibition of bile acid biosynthesis. Clin Gastroenterol Hepatol 2009;7:1189–1194. [PubMed: 19426836]
5. Camilleri M, Nurko S. Bile acid diarrhea in adults and adolescents. Neurogastroenterol Motil 2022;34:e14287. [PubMed: 34751982]
6. Harrison SA, Abdelmalek MF, Neff G, et al. Aldafermin in patients with non-alcoholic steatohepatitis (ALPINE 2/3): a randomised, double-blind, placebo-controlled, phase 2b trial. Lancet Gastroenterol Hepatol 2022;7:603–616. [PubMed: 35325622]
7. Hirschfield GM, Chazouillères O, Drenth JP, et al. Effect of NGM282, an FGF19 analogue, in primary sclerosing cholangitis: a multicenter, randomized, double-blind, placebo-controlled phase II trial. J Hepat 2019;70:483–493.
8. Harrison S, Abdelmalek M, Trotter J, et al. NGM282, a novel variant of FGF19, significantly reduces hepatic steatosis and key biomarkers of NASH: results of a Phase 2, multicenter, randomized, double-blinded, placebo controlled trial in biopsy-confirmed NASH patients. J Hepatol 2017;1:S92–S93.
9. Oduyebo I, Camilleri M, Nelson AD, et al. Effects of NGM282, an FGF19 variant, on colonic transit and bowel function in functional constipation: a randomized phase 2 trial. Am J Gastroenterol 2018;113:725–734. [PubMed: 29717197]
10. Rinella ME, Trotter JF, Abdelmalek MF, et al. Rosuvastatin improves the FGF19 analogue NGM282-associated lipid changes in patients with non-alcoholic steatohepatitis. J Hepatol 2019;70:735–744. [PubMed: 30529590]

Table 1.

Demographics, baseline characteristics, and effects of aldafermin and placebo in 30 patients with bile acid diarrhea.

Data median (IQR)	**Aldafermin (n=15)**	**Placebo (n=15)**	**P-value**
Demographics			
Age (years)	49.0 (40.0, 53.0)	43.0 (39.0, 58.0)	0.633
Sex (% Female)	80.0	93.3	0.282
BMI (kg/m^2)	31.5 (25.4, 36.7)	30.1 (23.2, 35.7)	0.561
Prior cholecystectomy, N (%)	6 (40.0)	5 (33.3)	0.705
Off BA sequestrants, N (%)	12 (80.0)	13 (86.7)	0.624
Baseline bowel function			
Stool consistency (BSFS 1–7)	5.3 (4.1, 5.6)	4.5 (4.2, 5.3)	0.407
Abdominal pain (0 to 10)	0.9 (0.0, 3.6)	0.4 (0, 1.0)	0.377
Stool frequency (BM/day)	2.3 (1.9, 3.2)	2.3 (1.8, 3.0)	0.901
Baseline bile acid synthesis and excretion			
Serum 7αC4 (ng/mL)	40.7 (17.5, 102.0)	53.0 (18.7, 82.4)	0.804
Serum FGF19 (pg/mL)	66.80 (34.2, 80.8)	94.30 (48.5, 108.8)	0.294
Random stool total BA (μmol/g)	3.1 (1.9, 4.0)	3.5 (1.9, 4.3)	0.913
% Primary BA (CA+CDCA)	5.4 (1.9, 14.6)	5.2 (2.0, 12.9)	0.896
% Secretory BA (DCA+CDCA)	63.8 (58.3, 66.7)	63.2 (57.4, 66.6)	0.777
On treatment bile acid synthesis and excretion *			
Serum 7αC4 (ng/mL), day 14	2.6 (1.6, 3.8)	50.5 (19.4, 122.0)	**<0.001**
Serum 7αC4 (ng/mL), day 28	2.2 (1.6, 3.3)	50.4 (25.8, 106.0)	**<0.001**
Random stool total BA (μmol/g), day 14	0.5 (0.3, 0.7)	2.7 (1.2, 5.1)	**0.002**
Random stool total BA (μmol/g), day 28	0.5 (0.3, 0.8)	3.6 (1.0, 7.3)	**<0.001**
% Primary BA (CA+CDCA), day 14	10.3 (3.3, 15.4)	2.8 (0.7, 9.8)	0.738
% Primary BA (CA+CDCA), day 28	9.9 (3.0, 20.0)	14.3 (7.4, 27.4)	0.417
% Secretory BA (DCA+CDCA), day 14	43.2 (32.5, 53.5)	59.2 (53.1, 63.9)	**<0.001**
% Secretory BA (DCA+CDCA), day 28	45.5 (39.8, 55.6)	60.0 (46.3, 65.1)	**0.033**
On treatment bowel function *			
Stool consistency, 1–28 days	4.7 (4.3, 5.1)	4.6 (4.4, 5.3)	0.224
Individual Δ stool consistency days 1–28 minus baseline^	−0.4 (-1.2, 0.1)	−0.02 (-0.4, 0.3)	0.125
Stool consistency, week 3	4.6 (4.3, 5.0)	4.8 (4.0, 5.3)	0.247
Stool consistency, week 4	4.4 (4.2, 5.0)	5.1 (4.4, 5.5)	0.047
Abdominal pain, 1–28 days	0.6 (0.0, 1.3)	0.5 (0.1, 1.1)	0.649
Stool frequency, 1–28 days	2.3 (2.1, 3.6)	2.1 (1.7, 2.7)	0.261

Bolded P-values are statistically significant. BA: bile acid, BM: bowel movements, BMI: body mass index, BSFS: Bristol Stool Form Scale, CA: cholic acid, CDCA: chenodeoxycholic acid, DCA: deoxycholic acid, FGF19: fibroblast growth factor-19, IQR: interquartile range, serum 7αC4: 7α-hydroxy-4-cholesten-3-one; Δ: change.

Stool consistency measured using BSFS 1–7 scale; Abdominal pain measured using 0–10 numerical scale; Stool frequency expressed as bowel movements (BM)/day. Stool consistency and frequency represent medians and interquartile ranges of the individual averages over the specified time periods.

[*] p-value obtained using ANCOVA adjusting for the baseline measurement of the outcome, BMI, sex, and baseline serum 7αC4;

[^] p-value obtained using Mann-Whitney Rank Sum test